

SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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COURSE CODE: CSA0712

COURSE NAME: COMPUTER NETWORK

1. DNS – Designing a Name Resolution Solution

a) Minimum set of DNS Resource Records :

The domain is **`univcampus.edu`**.

The following records are required:

Record	Name	Value	Purpose
A	`www.univcampus.edu`	`192.0.2.10`	Maps the web server name to its IPv4 address
A	`mail.univcampus.edu`	`192.0.2.20`	Maps the mail server name to its IPv4 address
A	`cse.univcampus.edu`	`192.0.2.30`	Maps the CSE departmental server to its IPv4 address
CNAME	`web.univcampus.edu`	`www.univcampus.edu`	Creates an alias for the web server
MX	`univcampus.edu`	`10 mail.univcampus.edu`	Specifies the mail server and its priority

DNS Zone Representation

dns

```
$ORIGIN univcampus.edu.

www     IN A      192.0.2.10
mail    IN A      192.0.2.20
cse     IN A      192.0.2.30

web     IN CNAME  www.univcampus.edu.

@       IN MX     10 mail.univcampus.edu.
```

Explanation:

- The **A record** maps a hostname directly to an IPv4 address.
- The **CNAME record** makes `web.univcampus.edu` an alias of `www.univcampus.edu`.
- The **MX record** tells mail clients that email for `univcampus.edu` should be delivered to `mail.univcampus.edu`.
- The priority **10** indicates the preference value of the mail exchanger.

b) DNS Resolution Steps for `web.univcampus.edu` :

When a client wants to access:

web.univcampus.edu

the following process occurs:

Step 1 – Client Query

The user's computer asks its configured **local DNS resolver**:

What is the IP address of web.univcampus.edu?

Step 2 – Resolver Checks Cache

The local resolver first checks its DNS cache.

- If the answer is cached and valid, it returns the cached result.
- Otherwise, it continues the resolution process.

Step 3 – Query Root DNS Server

The resolver queries a **root DNS server**.

The root server does not normally provide the final IP address. It directs the resolver toward the appropriate **`.edu` TLD DNS servers**.

Step 4 – Query `.edu` TLD Server

The resolver asks the `.edu` DNS server about:

univcampus.edu

The `.edu` server refers the resolver to the authoritative DNS server for:

univcampus.edu

Step 5 – Query Authoritative DNS Server

The resolver asks the authoritative server:

What is web.univcampus.edu?

The authoritative server returns:

web.univcampus.edu CNAME www.univcampus.edu

Step 6 – Resolve the CNAME

Because `web` is a CNAME, the resolver then looks up:

www.univcampus.edu

The authoritative server returns:

www.univcampus.edu A 192.0.2.10

Step 7 – Return Result to Client

The local resolver returns:

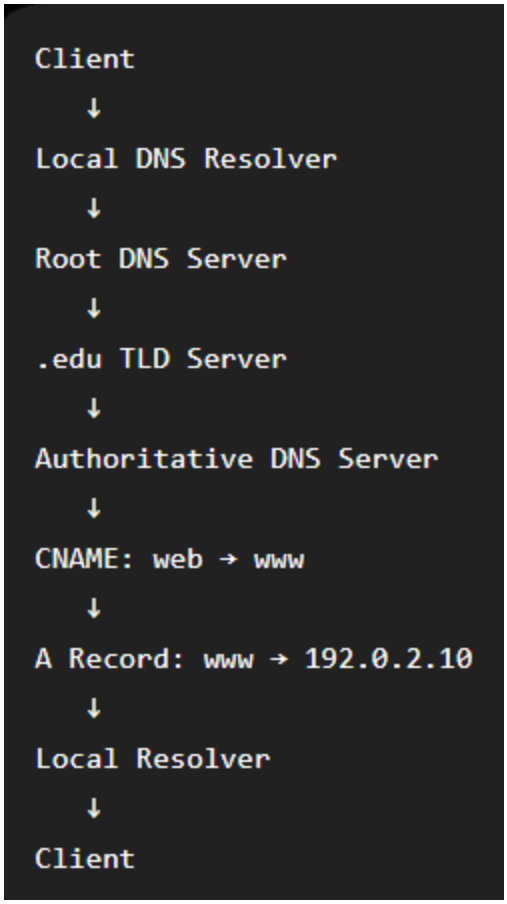
web.univcampus.edu → 192.0.2.10

Step 8 – Client Connects

The client can now connect to:

192.0.2.10

to access the web server.



c) Web Server IP Changes to `192.0.2.50` :

Initially:

www.univcampus.edu → 192.0.2.10

The new IP address is:

192.0.2.50

Record that must be updated

Only the **A record for** `www.univcampus.edu` needs to be changed:

www IN A 192.0.2.50

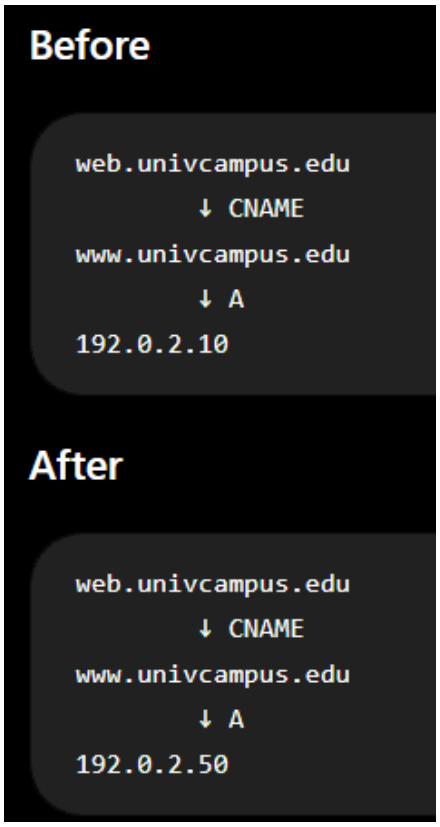
What about the CNAME?

The CNAME:

web IN CNAME www.univcampus.edu.

does not need to be changed.

This is because `web.univcampus.edu` does not directly contain the IP address. It points to `www.univcampus.edu`, which then points to the new IP.



The **MX record and other A records remain unchanged** because they are associated with different services.

Record	Update?	Reason
`www` A record	**Yes**	Web server IP has changed
`web` CNAME	**No**	It already points to `www`

`mail` A record	**No**	Mail server IP has not changed
`cse` A record	**No**	CSE server IP has not changed
MX record	**No**	Mail exchanger has not changed

Only the **A record of `www.univcampus.edu`** must be updated from `192.0.2.10` to `192.0.2.50`